

Lecture 2: Univariate Population Inference

July 30, 2023

Review



Outline



What do we mean by random draws?

i s	1 Y_1	2 Y_2	...	625 Y_{625}	$\bar{Y}_{625}$
1	1	1	...	0	.68
2	0	1	...	1	.70
3	0	0	...	1	.75
4	0	1	...	1	.73
5	0	1	...	1	.69
⋮	⋮	⋮	⋮	⋮	⋮
1 mil	0	1	...	1	.74
⋮	⋮	⋮	⋮	⋮	⋮
$f(\cdot)$	Be(.71)	Be(.71)		Be(.71)	$\frac{\text{Bin}(625, .71)}{625}$
E.(.)	.71	.71		.71	.71

Beyond binary

i	1	2	...	625	
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$
1	35	50	...	22	45
2	28	19	...	65	47
3	33	27	...	45	42
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	46
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(.)$	?	?		?	?
$E.(.)$	45	45		45	?

Important properties:

$$E[c] = c$$

$$E[aX] = aE[X]$$

Prove these assuming X is a discrete RV.

Higher moments about the mean. The j th *central* moment:

$$\mu_j = \text{E} \left[(X - \text{E}[X])^j \right] .$$

The 2nd *central* moment, the variance:

$$\begin{aligned} V(X) &= E \left[(X - E[X])^2 \right] \\ &= \\ &= \\ &= \\ &= \\ &= E \left[X^2 \right] - E[X]^2 \end{aligned}$$

- ▶ The average squared deviation from the expected value.
- ▶ Expectation of the square minus the square of the expectation.

What does this mean, physically?

The 2nd *central* moment, the variance:

$$\begin{aligned} V(X) &= E \left[(X - E[X])^2 \right] \\ &= E \left[X^2 - 2XE[X] + E[X]^2 \right] \\ &= E \left[X^2 \right] - 2E[XE[X]] + E \left[E[X]^2 \right] \\ &= E \left[X^2 \right] - 2E[X]E[X] + E[X]^2 \\ &= E \left[X^2 \right] - 2E[X]^2 + E[X]^2 \\ &= E \left[X^2 \right] - E[X]^2 \end{aligned}$$

- ▶ The average squared deviation from the expected value.
- ▶ Expectation of the square minus the square of the expectation.

What does this mean, physically?

What do we mean by random draws?

i	1	2	...	625	
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$
1	1	1	...	0	.68
2	0	1	...	1	.70
3	0	0	...	1	.75
4	0	1	...	1	.73
5	0	1	...	1	.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	0	1	...	1	.74
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	Be(.71)	Be(.71)		Be(.71)	$\frac{Bin(625, .71)}{625}$
E.(.)	.71	.71		.71	.71
V.(.)	.71(1-.71)	.71(1-.71)		.71(1-.71)	?

Beyond binary

i	1	2	...	625	
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$
1	35	50	...	22	45
2	28	19	...	65	47
3	33	27	...	45	42
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	46
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?		?	?
$V(\cdot)$	25	25		25	?

Important properties:

$$V[c] = 0$$

$$V[aX] = a^2 V[X]$$

$$V[c + aX] = a^2 V[X]$$

Prove these assuming X is a discrete RV.

What do these properties mean?

i	1	2	...	625	
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$
1	1	1	...	0	.68
2	0	1	...	1	.70
3	0	0	...	1	.75
4	0	1	...	1	.73
5	0	1	...	1	.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	0	1	...	1	.74
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	Be(.71)	Be(.71)		Be(.71)	$\frac{Bin(625, .71)}{625}$
E.(.)	.71	.71		.71	.71
V.(.)	.71(1-.71)	.71(1-.71)		.71(1-.71)	$\frac{625(.71)(1-.71)}{625^2}$

Definition:

Standard deviation: $\sigma(X) = \sqrt{V(X)}$.

Definition: mean squared error (MSE) about a guess c :

$$\begin{aligned} \mathbb{E} \left[(X - c)^2 \right] &= \\ &= \\ &= V(X) + (\mathbb{E}[X] - c)^2 \end{aligned}$$

Definition: mean squared error (MSE) about a guess c :

$$\begin{aligned} E \left[(X - c)^2 \right] &= E[X^2] - 2cE[X] + c^2 \\ &= E[X^2] - E[X]^2 + E[X]^2 - 2cE[X] + c^2 \\ &= V(X) + (E[X] - c)^2 \end{aligned}$$

What is the minimum MSE choice of c ?

Defining expectation over a vector of RVs.

$$E[(X, Y)] = (E[X], E[Y])$$

Defining an expectation over functions multiple R.V.s.

Let $Z = h(X, Y)$ be a scalar function of the random vector (X, Y) as we proceed.

Discrete:

$$E[Z] = \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} h(x, y) f(x, y)$$

Continuous:

$$E[Z] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dy dx$$

So this is just a straightforward generalization (working with joint PMFs/PDFs).

Important property: Prove it for discrete.

$$E[aX + bY + c] =$$

$$=$$
$$=$$
$$+$$

$$= aE[X] + bE[Y] + c$$

Important property: Prove it for discrete.

$$\begin{aligned} E[aX + bY + c] &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} h(x, y) f(x, y) \\ &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} (ax + by + c) f(x, y) \\ &= \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} a \cdot x \cdot f(x, y) \\ &\quad + \sum_{x \in \text{Supp}(X)} \sum_{y \in \text{Supp}(Y)} b \cdot y \cdot f(x, y) + c \\ &= aE[X] + bE[Y] + c \end{aligned}$$

What do these properties mean?

i	1	2	...	625	
s	Y_1	Y_2	...	Y_{625}	$\frac{1}{2}Y_1 + \frac{1}{2}Y_2$
1	35	50	...	22	$\frac{1}{2}35 + \frac{1}{2}50$
2	28	19	...	65	$\frac{1}{2}28 + \frac{1}{2}19$
3	33	27	...	45	$\frac{1}{2}33 + \frac{1}{2}27$
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	$\frac{1}{2}41 + \frac{1}{2}53$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?		?	?
$E(\cdot)$	45	45		45	$\frac{1}{2}45 + \frac{1}{2}45$

Understanding i.i.d.

What does i.i.d. stand for:

- ▶ Independent (i.e., observations independent from each other)
- ▶ Identically distributed (i.e., all observations drawn from same distribution)

Implications of independence

Y_1 and Y_2 are *independent* if and only if
 $f_{Y_1, Y_2}(y_1, y_2) = f_{Y_1}(y_1) \cdot f_{Y_2}(y_2)$, for all (y_1, y_2) .

If Y_1 and Y_2 are independent, then:

- Covariance is zero:

$$C(Y_1, Y_2) = 0.$$

- Variances are additive:

$$V(Y_1 + Y_2) = V(Y_1) + V(Y_2).$$

Implications of identically distributed

Y_1 and Y_2 are *independent* if and only if
 $f_{Y_1, Y_2}(y_1, y_2) = f_{Y_1}(y_1) \cdot f_{Y_2}(y_2)$, for all (y_1, y_2) .

Y_1 and Y_2 are *identically distributed* if and only if
 $f_{Y_1}(y_1) = f_{Y_2}(y_2)$ for all (y_1, y_2) .

So, we have n random variables that generate n values.

We can subscript each of these n random variables, as well as aggregate them.

The random vector $\mathbf{Y} = (Y_1, \dots, Y_n)'$.

Under the assumption of *random sampling* from F_Y , the collection of values that you see are each produced by an identical random process.

The actual values that we observe are denoted with $\mathbf{y} = (y_1, \dots, y_n)'$.

For now, we focus on learning about the population mean ($E[Y] = \mu$), although to do so, we will need to also learn about the nuisance parameter ($V[Y]$).

We want an estimator with small mean squared error (recall that this is equivalent to squared bias plus variance) and an estimator that allows the production of tight confidence intervals.

A sample statistic *summarizes* those n values. When used to estimate a population parameter, also considered an estimator.

Sample mean:

$$\hat{\mu}_{\text{sample mean}} = \bar{Y} = \frac{\sum_{i=1}^n Y_i}{n} = \frac{1}{n} (Y_1 + \dots + Y_n).$$

Class of linear combinations:

$$\hat{\mu}_{\mathbf{c}} = \sum_{i=1}^n c_i Y_i = (c_1 Y_1 + \dots + c_n Y_n).$$

Which column is the sampling distribution of the sample mean

i s	1 Y_1	2 Y_2	...	625 Y_{625}	$\bar{Y}_{625}$
1	35	50	...	22	45
2	28	19	...	65	47
3	33	27	...	45	42
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	46
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(.)$	?	?		?	?
$E.(.)$	45	45		45	?
$V.(.)$	25	25		25	?

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean, $E[\bar{Y}] = E[Y]$.

Proof:

$$\begin{aligned} E[\bar{Y}] &= E\left[\frac{1}{n}(Y_1 + \dots + Y_n)\right] \\ &= \\ &= \\ &= \\ &= \\ &= E[Y_1]. \end{aligned}$$

The sample mean is unbiased

i s	1 Y_1	2 Y_2	...	625 Y_{625}	$\bar{Y}_{625}$
1	35	50	...	22	45
2	28	19	...	65	47
3	33	27	...	45	42
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	46
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?		?	?
$E(\cdot)$	45	45		45	45
$V(\cdot)$	25	25		25	?

Within the class of linear combinations, what must be true for unbiasedness?

$$\hat{\mu}_{\mathbf{c}} = \sum_{i=1}^n c_i Y_i = (c_1 Y_1 + \dots + c_n Y_n) .$$

Property #2: The *sampling variance* of the sample mean,

$$V[\bar{Y}] = \frac{V[Y]}{n}.$$

Proof:

$$\begin{aligned} V[\bar{Y}] &= V \left[\left(\frac{1}{n} Y_1 + \dots + \frac{1}{n} Y_n \right) \right] \\ &= \\ &= \\ &= \\ &= \\ &= \frac{1}{n} V[Y]. \end{aligned}$$

Sampling distribution of the sample mean

i	1	2	...	625	
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$
1	35	50	...	22	45
2	28	19	...	65	47
3	33	27	...	45	42
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	46
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?		?	?
$E(\cdot)$	45	45		45	45
$V(\cdot)$	25	25		25	$\frac{25}{625}$

Within the class of linear combinations, what is the variance in terms of $\mathbf{c}$?

$$\hat{\mu}_{\mathbf{c}} = \sum_{i=1}^n c_i Y_i = (c_1 Y_1 + \dots + c_n Y_n).$$

What does this expression tell us about minimizing variance?

Learning about population variance

- ▶ We've shown that the sample mean is unbiased
- ▶ We've shown that the variance of the sample mean is $V[Y]/n$
- ▶ We still need to learn something about $V[Y]$

Recall that the “population” variance of a random variable Y :

$$V(Y) = E[Y^2] - E[Y]^2$$

Can we propose an estimator of $V(Y)$?

Put another way:

- ▶ Can we propose an estimator of $E[Y^2]$?
- ▶ Can we propose an estimator of $E[Y]^2$?

- ▶ Can we propose an estimator of $E[Y^2]$?

- ▶ Plug-in estimator:

$$\overline{Y^2} = \frac{\sum_{i=1}^n Y_i^2}{n}.$$

- ▶ $E[\overline{Y^2}] = E[Y^2]$ (just let $Y^2 = Z$ and it is just $E[\overline{Z}] = E[Z]$).

- ▶ Can we propose an estimator of $E[Y]^2$?
 - ▶ Plug-in estimator:

$$\bar{Y}^2 = \left[\frac{\sum_{i=1}^n Y_i}{n} \right]^2.$$

- ▶ What is its expectation ($E[\bar{Y}^2]$)?
- ▶ Recall that $V[\bar{Y}] = E[\bar{Y}^2] - E[\bar{Y}]^2$
- ▶ Simplifies to $\frac{V[Y]}{n} = E[\bar{Y}^2] - E[\bar{Y}]^2$
- ▶ Rearranging shows that $E[\bar{Y}^2] = E[\bar{Y}]^2 + \frac{V[Y]}{n}$

Therefore, the plug-in estimator of variance,

$$\overline{Y^2} - \overline{Y}^2.$$

is biased:

$$\begin{aligned} E[\overline{Y^2} - \overline{Y}^2] &= E[Y^2] - (E[Y]^2 + \frac{V[Y]}{n}) \\ &= (E[Y^2] - E[Y]^2) - \frac{V[Y]}{n} \\ &= V(Y) - \frac{V[Y]}{n} \\ &= (1 - \frac{1}{n})V(Y) \\ &= \frac{n-1}{n}V(Y) \end{aligned}$$

However, the bias can be corrected by multiplying the estimator by $\frac{n}{n-1}$:

$$\hat{V}(Y) = \frac{n}{n-1} \left[\overline{Y^2} - \overline{Y}^2 \right].$$

with

$$\blacktriangleright E[\hat{V}(Y)] = \frac{n}{n-1} E \left[\overline{Y^2} - \overline{Y}^2 \right] = \frac{n}{n-1} \frac{n-1}{n} V(Y)$$

This is what R computes when you type `var( . )`.

Sampling distribution of the sample variance (fill in for biased sample variance)

i	1	2	...	625			
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$	$\hat{V}(Y)$	$\overline{Y^2} - \bar{Y}^2$
1	35	50	...	22	45	20	
2	28	19	...	65	47	17	
3	33	27	...	45	42	38	
4	:	:	...	:	:		
5	:	:	...	:	:		
:	:	:	:	:	:		
:	:	:	:	:	:		
1 mil	41	53	...	21	46	32	
:	:	:	:	:	:		
:	:	:	:	:	:		
$f(\cdot)$	?	?		?	?		
$E(\cdot)$	45	45		45	45	25	
$V(\cdot)$	25	25		25	$\frac{25}{625}$		

Summary of “Small” Sample Properties

- ▶ $\text{MSE} = E[(\hat{\mu} - \mu)^2]$
- ▶ $\text{Bias} = E[(\hat{\mu} - \mu)]$
- ▶ $\text{Variance} = E[(\hat{\mu} - E[\hat{\mu}])^2]$

$$\underbrace{E[(\hat{\mu} - \mu)^2]}_{\text{MSE}} = \underbrace{E[(\hat{\mu} - \mu)]^2}_{\text{Bias squared}} + \underbrace{E[(\hat{\mu} - E[\hat{\mu}])^2]}_{\text{Variance}}$$

Oct. 4: Asymptotic (i.e., large sample) results and confidence intervals

Asymptotic properties of estimators

- ▶ Consistency (estimator converges in probability to the population parameter).
- ▶ $MSE \rightarrow 0$ (stronger than consistency, but more intuitive)
- ▶ Asymptotic normality (sampling distribution converges in distribution to normal distribution)

Property # 3: Given finite $E[Y]$ and finite $V(Y) > 0$, then for any $k > 0$,

$$\Pr(|\bar{Y} - E[Y]| \geq k) \leq \frac{V[Y]}{k^2 n}.$$

Proof follows directly from Chebyshev's inequality.

Property # 4. [Stated loosely.] *The sample mean is consistent for the population mean.* Given finite $E[Y]$ and finite $V(Y) > 0$, then as $n \rightarrow \infty$,

$$\Pr(|\bar{Y} - E[Y]| \geq k) \rightarrow 0.$$

Or, equivalently,

$$\bar{Y} \xrightarrow{p} E[Y],$$

where $\xrightarrow{p}$ is read as “convergence in probability.” This is a form of the *weak law of large numbers* (WLLN).

Consistency of $\overline{Y}$ for $E[Y]$

The WLLN states that

$$\overline{Y} \xrightarrow{P} E[Y],$$

If we think of $E[Y]$ as μ and $\overline{Y}$ as $\hat{\mu}$, then this is equivalent to saying that $\hat{\mu}$ is consistent for μ

Easier version based on MSE

$$MSE \rightarrow 0 \implies \hat{\theta} \rightarrow_p \theta$$

Implies that it is sufficient to show that Bias and Variance go to zero.

Consistency

With the person next to you, explain how you know that

$$\overline{Y} \rightarrow_p \mu$$

Another useful property

(A consequence of the) Continuous Mapping Theorem, stated loosely. Continuous functions are limit-preserving.

For continuous $f(\cdot)$, if $Y \xrightarrow{P} c$, then $f(Y) \xrightarrow{P} f(c)$.

Additionally, for continuous $f(b, c)$ if $X \xrightarrow{P} b$ and $Y \xrightarrow{P} c$, then $f(X, Y) \xrightarrow{P} f(b, c)$

Consistency

With the person next to you, are the following consistent?

- ▶ $\overline{Y^2} - \overline{Y}^2$ for σ^2
- ▶ $\hat{V}[Y] = \frac{n}{n-1}(\overline{Y^2} - \overline{Y}^2)$ for σ^2

Explain why?

Inconsistent Examples

With the person next to you, come up with at least one example of an estimator for μ that has the following properties:

- ▶ unbiased for μ but inconsistent
- ▶ variance goes to zero, but inconsistent for μ

Confidence Intervals

Important definition. The *standard error* of the sample mean:

$$\sqrt{V(\bar{Y})}.$$

(This is literally the standard deviation of the sampling distribution of the sample mean.)

An *estimate* of the standard error of the sample mean:

$$\sqrt{\hat{V}(\bar{Y})}.$$

Not unbiased (due to Jensen's inequality), though it is "consistent."

When you see "standard error" in a paper, this is what they mean (an estimate thereof). In theory, it is possible to estimate the sampling variance of the standard error of the sample mean, though.

What is the standard error of the sample mean and what is the estimated standard error of the sample mean?

i	1	2	...	625		
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$	$\hat{V}(Y)$
1	35	50	...	22	45	20
2	28	19	...	65	47	17
3	33	27	...	45	42	38
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$	
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
1 mil	41	53	...	21	46	32
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
$f(\cdot)$	?	?		?	?	
$E(\cdot)$	45	45		45	45	25
$V(\cdot)$	25	25		25	$\frac{25}{625}$	

It is therefore possible to estimate the average of a random variable (the population mean) and estimate our uncertainty (using an estimated SE).

However, we may want to know how likely it would be (across different random samples) for an interval of $\bar{Y} \pm 2$ SEs to contain $E[Y]$.

The central limit theorem provides an approach.

The central limit theorem (CLT) for the sample mean.

In words: if n is large, the sampling distribution of the sample mean ($\bar{Y}$) is approximately normal, regardless of the distribution of Y .

In sampling n times from any R.V. Y with finite $E[Y] = \mu$ and finite $V(Y) = \sigma^2$, then the standardized sample mean

$$Z = \sqrt{n}(\bar{Y} - \mu)/\sigma$$

converges in distribution as $n \rightarrow \infty$ to

$$\mathcal{N}(0, 1),$$

where $\mathcal{N}$ denotes the normal distribution.

I am not going to offer a proof here. The simplest “almost-proof” is given by the de Moivre–Laplace theorem (see the Wikipedia page).

A confidence interval for $E[Y]$, loosely speaking, is an *interval estimate* that covers the true $E[Y]$ with a given probability.

I.e., Suppose we were to repeatedly draw from $F_{\overline{Y}}$ and compute a (valid) 95% confidence interval for $E[Y]$. 95% percent of the time, this interval will encompass $E[Y]$.

Suppose we have:

- ▶ An asymptotically normal estimator of $E[Y]$: $\bar{Y}$
- ▶ A consistent estimator of $nV(\bar{Y})$: $n\hat{V}(\bar{Y})$

An approximate 95% confidence interval for $E[Y]$ would be:

$$\left(\bar{Y} - 1.96\sqrt{\hat{V}(\bar{Y})}, \bar{Y} + 1.96\sqrt{\hat{V}(\bar{Y})} \right).$$

An approximate 95% confidence interval for $E[Y]$ would be:

$$\left(\bar{Y} - 1.96\sqrt{\frac{\hat{V}(Y)}{n}}, \bar{Y} + 1.96\sqrt{\frac{\hat{V}(Y)}{n}} \right).$$

i	1	2	...	625		MoE	95% CI
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$	$1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$	$\bar{Y}_{625} \pm MoE$
1	35	50	...	22	45		
2	28	19	...	65	47		
3	33	27	...	45	42		
4	:	:	...	:	:		
5	:	:	...	:	:		
:	:	:	:	:	:		
1 mil	41	53	...	21	46		
:	:	:	:	:	:		
:	:	:	:	:	:		
$f.(.)$	?	?		?	?		
$E.(.)$	45	45		45	45		
$V.(.)$	25	25		25	$\frac{25}{625}$		

Table: Sampling Distributions for $\bar{Y}_{625}$, the Margin-of-Error, and the 95% Confidence Interval

i	1	2	...	625		MoE	95% CI
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$	$1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$	$\bar{Y}_{625} \pm MoE$
1	35	50	...	22	45	.198	
2	28	19	...	65	47	.195	
3	33	27	...	45	42	.204	
4	:	:	...	:	:	.201	
5	:	:	...	:	:	.205	
:	:	:	:	:	:	:	
1 mil	41	53	...	21	46	.197	
:	:	:	:	:	:	:	
:	:	:	:	:	:	:	
$f.(.)$	?	?		?	?		
$E.(.)$	45	45		45	45		
$V.(.)$	25	25		25	$\frac{25}{625}$		

Table: Sampling Distributions for $\bar{Y}_{625}$, the Margin-of-Error, and the 95% Confidence Interval

i	1	2	...	625		MoE	95% CI
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$	$1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$	$\bar{Y}_{625} \pm MoE$
1	35	50	...	22	45	.198	[44.83, 45.60]
2	28	19	...	65	47	.195	[44.46, 45.22]
3	33	27	...	45	42	.204	[44.61, 45.41]
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$	.201	[45.11, 45.88]
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$	.205	[44.54, 45.31]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	46	.197	[44.64, 45.46]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?		?	?		
E.(.)	45	45		45	45		
V.(.)	25	25		25	$\frac{25}{625}$		

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Suppose n is large.

Define the R.V.

$$Z = \frac{\bar{Y} - E[Y]}{\sqrt{V(\bar{Y})}},$$

where Z is distributed $\mathcal{N}(0, 1)$ (by asymptotic normality).

Plugging in the *estimated* standard error.

Define the R.V.

$$Z' = \frac{\bar{Y} - E[Y]}{\sqrt{V(\bar{Y})}} \frac{\sqrt{nV(\bar{Y})}}{\sqrt{n\hat{V}(\bar{Y})}} = \frac{\bar{Y} - E[Y]}{\sqrt{\hat{V}(\bar{Y})}}$$

where Z' is distributed $\mathcal{N}(0, 1)$ (by asymptotic normality, the consistency of variance estimator, and Slutsky's Theorem).

Bootstrapping

1. Take a sample of size n with replacement from your sample
2. Calculate $\mu^{(b=1)}$ from that bootstrap sample
3. Repeat for $b = 1, \dots, B$ where B is big
4. Do one of the following:
 - ▶ Calculate bootstrapped variance $\frac{1}{B-1} \sum_{b=1}^B (\mu^{(b)} - \overline{\mu^{(b)}})^2$
 - ▶ Put $\{\mu^{(b)} : b = 1, \dots, B\}$ in order and report the $\alpha/2$ and $1 - \alpha/2$ percentiles
 - ▶ ...

Go to bootstrapping code.

Why does bootstrapping work? (informal part 1)

- ▶ Empirical CDF ($\hat{F}(y) = \overline{I(Y \leq y)}$) is a collection of sample means/proportions for all y .

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- ▶ Therefore, the empirical CDF converges to population CDF ($F(y)$) for all y (AM pp. 116-117)
- ▶ If the empirical CDF is close enough to the population CDF, then a resample with replacement from the sample is like a sample from the population

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- ▶ And the plug-in estimators from resamples with replacement will converge toward a normal distribution in the same manner (with enough sample size).